

EDUCATION

Emory University • *Bachelor of Science in Mathematics and Computer Science* 4.0/4.0, 3x Dean's List May 2027

- Coursework: Data Structures and Algorithms, Machine Learning, Computer Vision, Database Systems, Computer Architecture, Linear Algebra, Multivariable Calculus

EXPERIENCE

LinkedIn | Software Engineering Intern, Azure Platform Tooling Team May 2026 – Present

- Built the recommendations agent, multi-agent orchestrator, and **LLM-as-judge** deterministic evaluation harness — extending toward **LangSmith**-based tracing — for a multi-agent Azure cost assistant that detects idle and underutilized resources, targeting **\$10M+** in identified annual savings across the org.
- Designed hard, config-level routing enforcement for the project's multi-agent orchestrator and shipped a platform-level bug fix to the shared eval runner, unblocking every other team's test suite org-wide.

Emory NLP/Naver | Researcher (collaboration with Naver Corporation) Dec 2025 – Present

- Designed a short-term user-memory (interest) representation for a personalization research project, and fine-tuned **Qwen3-14B (LoRA + groupwise DPO)** with a custom multi-signal reward function to extract it from user interaction logs.
- Diagnosed a reward-hacking failure mode causing the model to over-produce trivial single-evidence outputs, and redesigned the reward signal to fix it — recovering a verified improvement over baseline.

Lablup | Research/Software Engineering Intern Jun 2025 – Aug 2025

- Led GPU benchmarking and research automation, conducting **vLLM** performance analysis across Backend.AI's VM ecosystem to eliminate network-latency variables, verified performance consistency, and built an auto-benchmarking pipeline with per-request token throughput metrics.

Nell Hodgson Woodruff School of Nursing – MAIX Lab | AI Researcher May 2025 – Dec 2025

- Advanced multimodal ECG representation learning by integrating concept-based dictionaries into clinical report text embeddings, improving signal-report cross-modal alignment accuracy by 9% over the **MERL** approach.
- Applied **FAISS** for scalable disease-concept retrieval, reducing retrieval latency by 65% over lexical methods.

Sionic AI | ML Research Intern May 2024 – Jan 2025

- Researched **hard negative mining** (HNM) strategies to improve contrastive learning for the company's **RAG** pipeline.
- Boosted BGE-M3 fine-tuning by 4.87 points on Korean MTEB retrieval task with a positive-aware HNM strategy.
- Fine-tuned pretrained models with **PyTorch**, achieving an 85.73 BLEU in Korean grammar error correction, outperforming SOTA models GPT-4 (75.03) and HCX-3 (71.24); published findings at PACLIC 38.

PROJECTS

Curator: AI-Powered Coursework Management Tool • [GitHub](#) • [YouTube](#) • [Devpost](#)

- A highly concurrent automated local file organizer and processor as **Go** backend, using the **Actor Model** with goroutines and channels, **Gemini Go SDK**, and mutex-protected caching with JSON-backed models for persistence.
- Expanded with **Next.js/React** frontend with **Typescript** and **Tailwind CSS** to display organized course materials, up-to-date quizzes and study guides, and interactive knowledge graphs across courses.

MBTI Dating Simulation App • [GitHub](#)

- AI dating simulation app using a **Langflow**-based two-stage **RAG** pipeline to generate MBTI-personalized conversations, with dynamic, context-aware dialogue combining embedded personality traits with conversational memory.
- Built with Python, Streamlit, Langflow, Claude API, Cohere, Pinecone, and Stable Diffusion.

Good Hangul • [goodhangul.sionic.ai](#)

- On-premise Korean grammar correction platform powered by finetuned nllb-200-ko-gec-3.3B model hosted on **Amazon S3**, **Flask** API backend, **PocketBase** DB for collecting training data, and **Next.js/React/TypeScript** web interface.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Project Emory | President Aug 2023 – Present

- Manage 3 project teams of 20+ Emory student engineers building software for on-campus initiatives in collaboration with faculty.

Emory Hacks | Co-Founder Aug 2023 – Present

- Co-founded Emory's first MLH-certified hackathon engaging 200+ participants; led the sponsorship team, securing **\$20K+** from firms like Hyundai, JPMorgan, and Wolfram Alpha.

SKILLS

Languages: Python, Go, Java, JavaScript/Typescript, SQL, C, HTML, CSS, Shell Script

ML/AI: PyTorch, LoRA, DPO, RAG, FAISS, Hugging Face, Qwen3, LLM-as-judge evaluation, Pandas, Claude SDK, GitHub Copilot

Frameworks/Libs: React, Next.js, Node.js, Django, FastAPI, GraphQL, Nginx, PostgreSQL

Tools: Git, Linux, Docker, AWS (S3, EC2, Amplify), GCP (Cloud Run/Build, Gemini), Azure, Jira, Gemini SDK